

Population genetic structure of the endangered Butler's Gartersnake (*Thamnophis butleri*): does the Short-headed Gartersnake (*Thamnophis brachystoma*) exist in Canada?

Daniel W.A. Noble, Jonathan D. Choquette, John S. Placyk, Jr., and Ronald J. Brooks

Abstract: Understanding population genetic structure is fundamental to conservation of endangered species. It is particularly important when working with species that are morphologically conserved because strong genetic divisions could represent cryptic species. Butler's Gartersnake (*Thamnophis butleri* (Cope, 1889)) is an endangered species in Canada, having a fragmented distribution and being restricted to southwestern Ontario. Furthermore, it is difficult to distinguish morphologically from a closely related species, the Short-headed Gartersnake (*Thamnophis brachystoma* (Cope, 1892)). We use mitochondrial DNA (mtDNA) and seven microsatellite DNA loci to evaluate the genetic structure of Canadian *T. butleri* populations and to test for the presence of *T. brachystoma* in one of these populations. All individuals had the same mtDNA haplotype, and there was no evidence of multiple, syntopic genetic clusters, thereby rejecting the hypothesis that *T. butleri* and *T. brachystoma* co-exist in Canada. Two different model-based assignment tests using microsatellite DNA data suggest that there are four to five genetically distinct clusters of *T. butleri* (F_{ST} from 0.12 to 0.20). We provide the first population genetic study of *T. butleri* in Canada and refute the presence of *T. brachystoma*. Our results may provide guidance on recovery strategies for this species and identify areas to target fine-scale genetic analyses.

Key words: assignment test, Butler's Gartersnake, conservation genetics, population genetics, Ontario, Short-headed Gartersnake, *Thamnophis brachystoma*, *Thamnophis butleri*.

Résumé : La compréhension de la structure génétique des populations est un aspect fondamental de la conservation des espèces en voie de disparition et est particulièrement importante dans le cas d'espèces morphologiquement conservées étant donné que de fortes divisions génétiques peuvent représenter des espèces cryptiques. La couleuvre à petite tête (*Thamnophis butleri* (Cope, 1889)) est une espèce en voie de disparition au Canada, son répartition, limitée au sud-ouest de l'Ontario, étant fragmentée. Il est en outre difficile de la distinguer, sur une base morphologique, de *Thamnophis brachystoma* (Cope, 1892), une espèce étroitement apparentée. Nous utilisons l'ADN mitochondrial (ADNmt) et sept microsatellites d'ADN pour évaluer la structure génétique des populations canadiennes de *T. butleri* et vérifier la présence de *T. brachystoma* dans une de ces populations. Tous les individus présentent le même haplotype d'ADNmt, et rien n'indique la présence de multiples groupements génétiques syntopiques, ce qui invalide l'hypothèse de la coexistence de *T. butleri* et *T. brachystoma* au Canada. Deux tests d'affectation différents reposant sur des modèles et faisant appel aux données de microsatellites d'ADN laissent croire à la présence de quatre à cinq groupements génétiques distincts de *T. butleri* (F_{ST} de 0,12 à 0,20). Nous présentons la première étude de la génétique des populations de la couleuvre à petite tête au Canada et démontrons l'absence de la couleuvre *T. brachystoma*. Nos résultats pourraient servir à orienter les stratégies visant le rétablissement de *T. butleri* et à cerner les zones à cibler des analyses génétiques fines. [Traduit par la Rédaction]

Mots-clés : test d'affectation, couleuvre à petite tête, génétique de la conservation, génétique des populations, Ontario, *Thamnophis brachystoma*, *Thamnophis butleri*.

Introduction

Determining the number of distinct populations and evaluating their genetic diversity is fundamental to the effective design of conservation strategies for threatened and endangered species (Avice 1992; Haig 1998; Crandall et al. 2000; Hedrick 2001; DeYoung and Honeycutt 2005). This information can be used to focus resources on conserving populations that (i) have high genetic diversity, (ii) have unique genetic structure, or (iii) may be in need of increased genetic diversity via the addition of more individuals by translocation or from captive-breeding efforts (Avice 1992; Crandall et al. 2000). Distinct genetic structure may also

represent evolutionarily significant variation within a species, or it may represent an unknown or undescribed species (Haig 1998; DeYoung and Honeycutt 2005; Bickford et al. 2006). Unfortunately, the patchy distributions, cryptic behaviour, and low densities of many endangered species make it difficult to obtain adequate sample sizes for population genetic analyses. However, recent advances in population genetics and statistical inference have alleviated the need for large sample sizes from single populations, making analysis of population structure more feasible (Manel et al. 2005) when constrained by small samples.

Model-based assignment tests have become popular tools and have addressed a diverse array of questions including character-

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ization of cryptic species (Noble et al. 2010) and understanding population genetic structure (Manel et al. 2005; Crosby et al. 2009). Assignment methods use multilocus genotypes to assign individuals to clusters or groups that are close to Hardy–Weinberg and linkage equilibrium (Manel et al. 2005). The identified clusters are meant to represent populations of interbreeding individuals. Such analyses are ideal for species that are difficult to collect or are not very abundant because they use individual rather than population-level information. Their application to species of conservation concern has been particularly fruitful and has provided insight into management units where conservation efforts should be directed. For example, Tzika et al. (2008) used assignment methods to delineate populations of the endangered Jamaican Boa (*Epicrates subflavus* Stejneger, 1901). Combining mitochondrial and microsatellite DNA markers, they successfully identified three major nuclear clusters that were genetically differentiated from one another. Haplotype networks suggest that two of the identified nuclear genetic clusters correspond to an eastern and a western + central haplogroup, which suggested that these units should be managed separately (Tzika et al. 2008). Studies such as these highlight the importance of these tools in developing more informed management plans for threatened or endangered species.

Butler's Gartersnake (*Thamnophis butleri* (Cope, 1889)) is a small (total length: 38–51 cm; Conant and Collins 1991), secretive snake that is patchily distributed in remnant prairies, cultivated meadows, and naturalized urban habitats (Planck and Planck 1977; COSEWIC 2011). It is a dietary specialist, feeding almost exclusively on earthworms and leeches throughout the active season, which is generally from April to September (Carpenter 1952; Casbourn et al. 1976). During this period, it spends most of its time foraging in long grass and exhibits reduced activity in the drier months of July and August (Planck and Planck 1977). *Thamnophis butleri* is restricted to the Great Lakes region of North America, and in Canada, it is found only in southwestern Ontario. *Thamnophis butleri* was recently listed as Endangered in Canada because of the loss of historical populations and ongoing threats to their habitat (COSEWIC 2011). Here we focus on three distinct regions within the Canadian range of this species (Windsor, Sarnia, and Luther Marsh; Fig. 1) that encompass almost all of their known distribution in Canada. The Windsor and Sarnia populations are separated from the United States populations by the Detroit River and Lake St. Clair, respectively. The Luther Marsh population is the most northern extent of *T. butleri*'s distribution and is geographically isolated (~250 km) from populations in both Canada and the United States.

Within one of these regions (Luther Marsh), investigators have reported *T. butleri* with reduced numbers of dorsal and labial scales that either approach or conform to counts characteristic of the closely related Short-headed Gartersnake, *Thamnophis brachystoma* (Cope, 1892) (Schueler and Westell 1975; Sandilands 1984; Oldham and Sutherland 1986). *Thamnophis brachystoma* is a smaller relative of *T. butleri* and has only been confirmed in New York and Pennsylvania, USA. *Thamnophis brachystoma* is described as having dorsal scale counts in a 17–17–17 or a 17–17–15 (anterior dorsal to mid-dorsal to posterior dorsal) arrangement and six upper labials, whereas *T. butleri* is described as having dorsal scale counts in a 19–19–19 or 19–19–17 arrangement and seven upper labials (Smith 1945; Conant 1950). Placyk et al. (2012) compared mitochondrial gene sequences from the ND2 gene region across the Plains Gartersnake (*Thamnophis radix* (Baird and Girard, 1853)), *T. butleri*, and *T. brachystoma* in the USA and found that *T. brachystoma* is genetically divergent in mtDNA from *T. butleri*; however, this is the only genetic comparison between these two allopatric species to date.

Investigators have proposed that the Canadian *T. butleri* from Luther Marsh with reduced scale counts may be examples of intergradation (hybridization) between *T. butleri* and *T. brachystoma*

(Schueler and Westell 1975; Planck and Planck 1977; Sandilands 1984; Oldham and Sutherland 1986; Harding 1997; Tennant and Bartlett 2000), or are in fact *T. brachystoma* (Coulson and Peluch 1984). Conversely, others have demonstrated that overlap in both dorsal and labial scale counts is occasionally observed between these two species (Ruthven 1908; Logier 1939; Smith 1945, 1949; Planck and Planck 1977), leading one to question the significance of reduced scale counts in Luther Marsh and whether they are within the normal extremes for *T. butleri* and not evidence for hybridization or the presence of *T. brachystoma* in Canada. Nonetheless, based on the central geographic position of Luther Marsh relative to the known range of both *T. brachystoma* and *T. butleri* and the presence of hybridization between *T. butleri* and other *Thamnophis* elsewhere (*T. radix*: Fitzpatrick et al. 2008), it seems plausible to hypothesize a hybrid zone at Luther Marsh. However, to date no genetic studies have been conducted to verify the existence of *T. brachystoma* in Canada. If substantiated, *T. brachystoma* would be the newest edition to the list of Canadian reptiles and require immediate management to ensure their protection.

Here we characterize the population genetic structure of *T. butleri* in Canada using spatially implicit and explicit Bayesian model-based assignment tests to address two questions. First, we try to resolve a long-standing question by evaluating whether variable diagnostic traits found in one population (Luther Marsh; Fig. 1) represent *T. brachystoma*, and second, we test whether there is significant population genetic structure in Canada. More specifically, we address the following questions. (i) Does *T. brachystoma* co-occur with *T. butleri* in the Luther Marsh population? (ii) How many genetically distinct populations exist within the Canadian range of this species? We combine both maternally inherited mitochondrial DNA (mtDNA) and seven nuclear microsatellite DNA loci to test these two questions. Finally, we discuss the implications of our results for the conservation of *T. butleri* in Canada.

Materials and methods

Sample collection

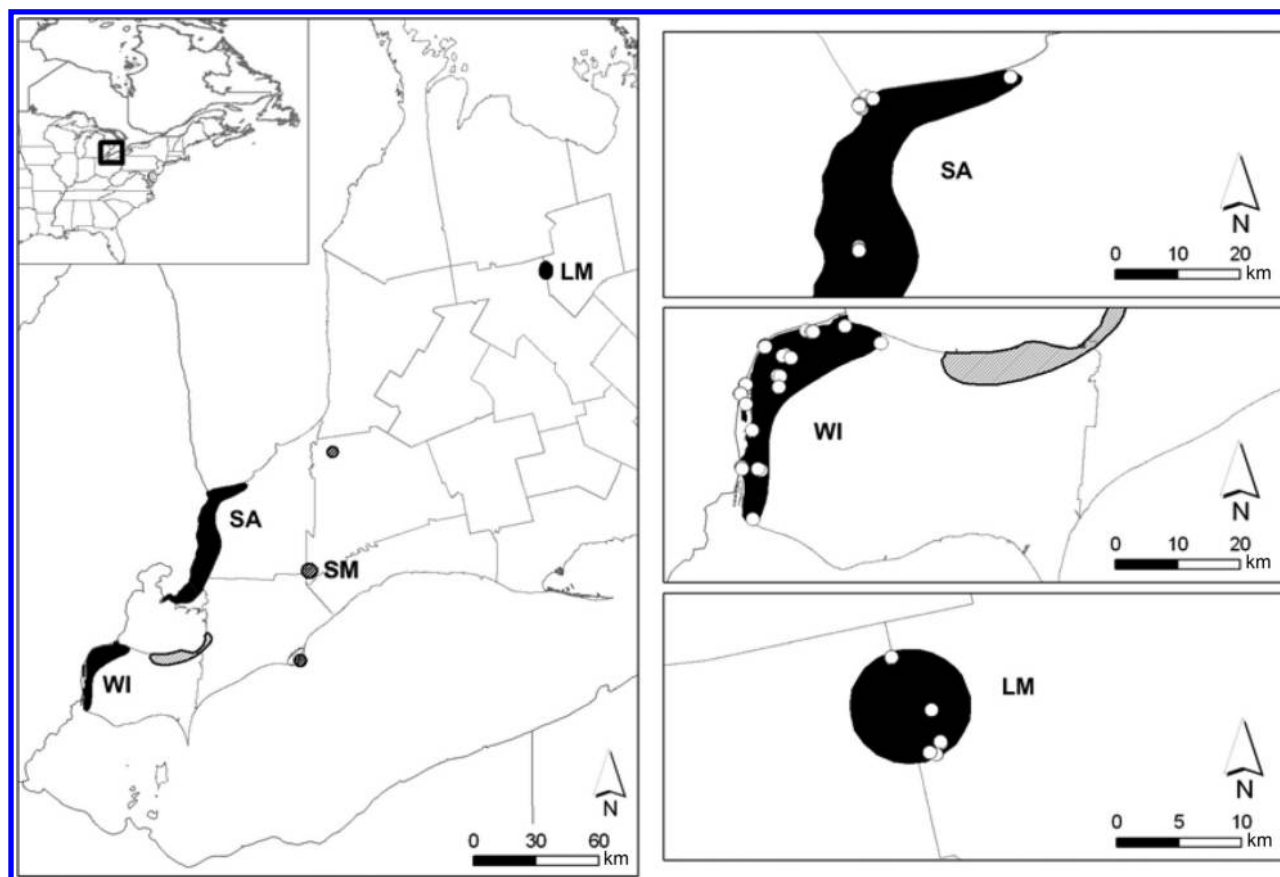
A total of 112 *T. butleri* were sampled from numerous locations within the Windsor area (Essex Co.: 42.2°N, 83.0°W), Sarnia area (Lambton Co.: 43.0°N, 82.4°W), and at the Luther Marsh Conservation Area (Wellington Co. and Dufferin Co.: 44.0°N, 80.4°W) between 28 April and 11 June 2009. This was the most extensive survey of this species in Ontario since 1976 (Planck and Planck 1977). There are historical records of *T. butleri* from locations in southwestern Ontario that were not sampled by us (Fig. 1); however, it is probable that the species has since become extirpated from some or all of these areas (COSEWIC 2011). For example, *T. butleri* was first recorded in Canada from Skunk's Misery (Fig. 1; Logier 1939), but has not been reported from this area since 1981 (COSEWIC 2011; F.W. Schueler, personal communication, 2009). Multiple searches by the authors in 2009 at many historical locations, including Skunk's Misery, did not yield any specimens, and as a result, data from these sites were not available for analyses.

Snakes were measured for snout–vent length (SVL) and body mass (BM), and were aged and sexed (adults) visually based on a combination of size, girth, tail width, and tail length (Planck and Planck 1977). A few millimetres of tail tissue was excised from each snake (total $n = 112$ —Windsor, $n = 60$; Sarnia, $n = 27$; Luther Marsh, $n = 25$) and stored in 95% ethanol for genetic analyses. We counted dorsal scales (anterior, mid-dorsal, posterior) on most individuals from the Windsor area ($n = 40$) and Sarnia area ($n = 21$). At Luther Marsh, we counted dorsal scales plus upper labial and lower labial scales on most individuals ($n = 24$).

Mitochondrial DNA

Genomic DNA was extracted using standard phenol–chloroform–isoamly extraction procedures (Sambrook et al. 1989). We successfully amplified an ~800 base-pair fragment of the NADH

Fig. 1. Current and historical distribution of Butler's Gartersnake (*Thamnophis butleri*) in Canada (left panel). Black areas indicate areas where *T. butleri* are still known to be present (LM, Luther Marsh Conservation Area: 44.0°N, 80.4°W; SA, Sarnia area: 43.0°N, 82.4°W; WI, Windsor area: 42.2°N, 83.0°W). Grey-hatched areas indicate areas with historical records of *T. butleri* (SM, Skunk's Misery: 42.6°N, 81.8°W). Collection sites within each region are represented by white circles (right panel).



dehydrogenase subunit 2 (ND2) mitochondrial gene from 90 individuals distributed in Windsor, Sarnia, and Luther Marsh (Fig. 1) using the primers L4437b (5'-CAG CTA AAA AAG CTA TCG GGC CCA TAC C-3') and Sn-ND2r (5'-GGC TTT GAA GGC TMC TAG TTT-3') (Kumazawa et al. 1996; Placyk et al. 2007). Polymerase chain reactions (PCRs) were conducted in 25 μ L volumes with 1.0 μ L DNA template, 1.0 μ L of each primer (10 pmol/ μ L), 2.0 μ L dNTP (10 mmol/L of each dNTP; Roche Diagnostics), 1.5 μ L Mg^{2+} (25 mmol/L), 2.5 μ L 10 $\times$ loading buffer (Takara), and 0.25 μ L *Taq* polymerase (Takara). We modified reaction conditions for samples that did not amplify by adding 12 μ L of polyethylene glycol and 1 μ L of trehalose to PCRs. The addition of these macromolecules increased reaction efficiency by improving the interaction between reaction components and *Taq* polymerase (Zimmerman and Harrison 1987; Spiess et al. 2004; Zhang et al. 2009). Thermal cycling was performed with an initial denaturation at 95 $^{\circ}$ C for 5 min followed by 30 cycles of 94 $^{\circ}$ C for 1 min, 55 $^{\circ}$ C annealing for 1 min, 72 $^{\circ}$ C for 1.5 min, and a final extension of 72 $^{\circ}$ C for 5 min. All PCR products were run on 1% agarose gels and purified using QIAquick PCR purification kits (Qiagen). The cleaned products were directly cycle-sequenced in the forward direction using the L4437b primer. All DNA sequencing reactions were performed using Big-Dye terminator sequencing chemistry with an ABI 3730 sequencing machine (Applied Biosystems). Sequences were visualized, corrected, and aligned using SEQUENCHER version 4.5 (Genecode Corp). The sequences in this study were also utilized by Placyk et al. (2012) in a broader, range-wide biogeographic study of *T. butleri*.

Microsatellite DNA

Seven microsatellite DNA loci were successfully amplified for 103 individuals (Table 1). The DNA template was purified using columns prior to all PCR reactions to resolve amplification problems (Qiagen). Reactions took place in 10 μ L volumes with 2.0 μ L DNA template, 0.5 μ L of each primer (10 pmol/ μ L), 0.2 μ L dNTP (10 mmol/L of each dNTP; Roche Diagnostics), locus-specific $MgCl_2^{2+}$ (Table 1), 1.0 μ L 10 $\times$ loading buffer (Takara), and 0.1 μ L of *Taq* polymerase (Takara). All forward primers were labelled with tetrachloro-6-carboxy-fluorescein (TET). Reactions took place in a PTC-200 thermocycler (MJ Research) with an initial denaturation of 94 $^{\circ}$ C for 2 min, followed by 30–36 cycles of 94 $^{\circ}$ C for 1 min, primer-specific annealing temperature (Table 1) for 1 min, and 72 $^{\circ}$ C for 1 min. All reactions had a final extension of 72 $^{\circ}$ C for 2 min. PCR products were run with a GenescanTM-350 TAMRA size standard (Applied Biosystems) and a marker individual and separated on 6% polyacrylamide gels and visualized with an FMBIO laser scanner (Hitachi). A set of 24 individuals on each locus were rerun and scored again to ensure that alleles were scored the same across gels.

Data analysis

We used Bayesian model-based assignment tests to test for the presence of *T. brachystoma* and assess the degree of genetic structure in Canada, particularly in the Luther Marsh population (Fig. 1). These methods use multilocus genotypes and a predefined number of clusters (*K*) to generate clusters that minimize deviation from Hardy–Weinberg equilibrium (HWE) and linkage

Table 1. Microsatellite DNA loci that were successfully amplified in Butler's Gartersnake (*Thamnophis butleri*) from southern Ontario.

Locus	Forward sequence (5'–3')	Reverse sequence (5'–3')	Repeat	T _{anneal} (°C)	MgCl ₂ ⁺ (μL)	Fragment size (range)	No. of alleles
Te1Ca3*	F: CCCCCACCTACCTACCTG	R: TGGGTAGGGCAAAAACAG	(CA) ₁₃	54	0.3	88 (74–94)	13
Te1Ca18*	F: CCCCACATTTTGGCAAG	R: TCGGAGTGTGTGGAGTG	(CA) ₂₂ (GA) ₂ (C) ₆	54	0.3	129 (91–127)	10
Te1Ca29*	F: TGCCTTATTTGCTTGGGTTG	R: TCTTTCAACCTGCTTTGTAGACAC	(CA) ₅ (N) ₄ (CA) ₄ ; (CG) ₁ (CA) ₁	55	0.6	154 (156–190)	10
TS010†	F: TGACTCAGATGCCCTCAGTCTA	R: CGGACCAACCAGGAACAGAAAT	(ATGG) ₃ (ATGA) ₆	63	0.6	108–146	8
TS2‡	F: GGCTAGCCCCGTGTCTCTT	R: CACAAC'TCCAAATATTGAAGATTA	(AAT) ₁₀ + (AAT) ₃	56	0.6	118–156	15
Nsμ2§	F: TCCTCTTTGGCAGATAATAGT	R: AGCCGAGAACACACTAGTAAGT	(AC) ₁₈	55	0.6	144–176	6
Nsμ3§	F: CTGACTCACTTCTGACCTAAT	R: AATATTTGCTTGGCTCAAAC	(ATCT) ₁₄ + AT(CA) ₂₀	55	0.6	157–189	16

Note: Loci information, annealing temperatures (T_{anneal}), MgCl₂⁺ added, base-pair fragment range, and number of alleles are provided.

*Garner et al. 2004.

†Manier and Arnold 2005.

‡McCracken et al. 1999.

§Prosser et al. 1999.

equilibrium (Manel et al. 2005). Individuals are then assigned probabilistically to one or more cluster(s) based on their multi-locus genotype. If two species exist in the Luther Marsh population, then we would predict two distinct nuclear DNA clusters with disparate mtDNA haplotypes living syntopically (Noble et al. 2010).

Admixture and nonadmixture models were run in STRUCTURE version 2.2 (Pritchard et al. 2007) and a spatial admixture model in GENELAND version 3.2.4 (Guillot et al. 2005b) varying the parameter *K* (number of a priori populations) between 1 and 15. In STRUCTURE, we ran 60 000 iterations and designated 10 000 runs as burn-in. This was repeated 100 times for each value of *K* and the 10 runs with the highest log-likelihood (Ln P(D)) were averaged and plotted to determine the most likely value of *K*. In GENELAND, we ran a spatial model with correlated allele frequencies with 50 independent runs each with 100 000 iterations and a burn-in of 200 saved iterations (thinning = 100) while varying *K* between 1 and 15. We adopted a similar approach to inferring *K* as that used by Coulon et al. (2006), by running 50 independent runs and choosing the mode of these 50 runs as the most likely value of *K*. We then carried out 50 independent runs with *K* fixed at this modal value as suggested by Guillot et al. (2005a). Recent changes to the correlated allele frequency model have made it possible to estimate *K* in a single step (Guillot 2008); however, we found that the initial run assigned individuals in the data set with very low probabilities to the identified clusters. A second run with *K* fixed at the modal value gave much more consistent results and higher individual probabilities of cluster membership. We chose the five runs of the modal *K* value with the highest posterior probability and we compared the spatial clustering across these five runs for consistent spatial clustering patterns. Individual assignment probabilities were plotted spatially and superimposed on a map of North America using the statistical software package R (R Development Core Team 2010).

We obtained measures of heterozygosity, the number of private alleles, and allelic richness using ARLEQUIN version 3.5 (Excoffier et al. 2005) and ANDZ version 1.0 (Szpiech et al. 2008). We conducted these analyses on three geographically disparate regions (Windsor region, Sarnia region, and Luther Marsh). We limited our analyses to the latter three regions because different genetic clusters identified within the Windsor region consisted of too few individuals on their own to obtain meaningful estimates of allele frequencies, heterozygosity, and allelic richness, and because it was not clear exactly how many genetic clusters existed in the Windsor region based on the results of both assignment programs.

Results

Morphological and genetic diversity in Luther Marsh

Thamnophis butleri from the Luther Marsh population exhibited reduced scale counts (Table 2); however, no individuals were ob-

served with typical *T. brachystoma* scalation (dorsal scales in a 17–17–15 or 17–17–17 arrangement and six left and right upper labials). Of the 24 specimens that we measured, 18 (75%) expressed a reduction in either the number of dorsal scales or the number of upper labial scales (Table 2). Furthermore, 8 out of 24 specimens (33%) expressed a reduction in dorsal scalation (including one 17–17–17 snake), while 6 out of 24 specimens (25%) expressed a reduction in both the number of dorsal and upper labial scales (Table 2). One third of Luther Marsh snakes had an anterior dorsal scale count less than 19 and no snakes had an anterior dorsal scale count greater than 19, whereas in Sarnia, no snake had fewer than 19 (Table 2). Greater variability in anterior dorsal scale counts was witnessed in snakes from the Windsor area (Table 2). The mean (±SD) anterior dorsal scale count from Luther Marsh snakes (18 ± 0.9) is statistically inferior to that measured from Sarnia snakes (20 ± 0.9, Mann–Whitney test, *p* < 0.01) and Windsor snakes (19 ± 0.8, *p* < 0.05).

All snakes that we collected, including specimens in Luther Marsh, contained a single mtDNA haplotype with no nucleotide variation. This haplotype was shared with *T. butleri* from Michigan and Ohio (Placyk et al. 2012). Snakes from Luther Marsh contained 12 private alleles at 5 out of 7 microsatellite DNA loci (*Te1Ca18*, *Te1Ca3*, *Te1Ca29*, *TS2*, and *Ns3*). *Te1Ca18* had a single private allele, while *Te1Ca3*, *Te1Ca29*, and *TS2* each had two private alleles. Locus *Ns3* had five private alleles. The Luther Marsh population had an *H_O* = 0.67 ± 0.05 (mean ± SE) and an *H_E* = 0.69 ± 0.03 and did not deviate significantly from HWE and linkage equilibrium. STRUCTURE and GENELAND grouped all individuals from the Luther Marsh population into a single, genetically divergent (*F_{ST}* = 0.12–0.20) cluster irrespective of the value of *K* (Figs. 3a, 3b; Table 3). None of the individuals with morphology approaching *T. brachystoma* grouped into a genetically distinct cluster at any value of *K*.

Allelic diversity and the number of genetically distinct populations

Allelic diversity among the seven microsatellite DNA loci ranged from 6 to 16 alleles, with a mean of 11.14 alleles/locus (Table 1). No loci were in linkage disequilibrium after Bonferroni correction (*α* = 0.002). Locus *TS2* deviated significantly from HWE in snakes from Luther Marsh and the Windsor region. Snakes from Luther Marsh had a lower *H_O* (0.67) than *H_E* (0.69) and those from the Windsor region had a higher *H_O* (0.69) than *H_E* (0.77). Windsor snakes also deviated significantly from HWE at loci *TS10* and *Te1Ca18* and in both cases these loci had heterozygote deficiencies. Snakes from the Sarnia region also had a significant heterozygote deficiency at locus *Te1Ca29*. Although these could indicate the presence of null alleles, the loci in disequilibrium were not consistent across all three populations and so we retained all loci for further analyses. Windsor snakes had a higher mean number of alleles and heterozygosity than snakes from the

Table 2. Dorsal and labial scale counts from 85 Butler's Gartersnakes (*Thamnophis butleri*) sampled from three regions (Luther Marsh, Sarnia, and Windsor) in Ontario, Canada, in 2009.

Population	N	Dorsal scale				Upper labial scale		Lower labial scale		Percentage of snakes with the following scale counts					
		Anterior	Middle	Posterior		Right	Left	Right	Left	ADS > 19	ADS < 19	U/S (left or right) < 7	U/S (both sides) < 7 and ADS < 19	U/S (left or right) < 7 and ADS < 19	U/S (left or right) < 7 or ADS < 19
Luther Marsh	24	18±0.9	19±0.4	17±0.2		6±0.5	6±0.6	7±0.4	7±0.4	0	33	67	42	17	75
Sarnia	21	20±0.9	19±0.8	17±0.6		NA	NA	NA	NA	29	0	NA	NA	NA	NA
Windsor	40	19±0.8	19±0.0	17±0.2		NA	NA	NA	NA	13	10	NA	NA	NA	NA

Note: The Short-headed Gartersnake (*Thamnophis brachystoma*) is described as having dorsal scale counts in a 17–17–17 or a 17–17–15 (anterior to mid-dorsal to posterior) arrangement and six upper labials, whereas *T. butleri* is described as having dorsal scale counts in a 19–19–19 or 19–19–17 arrangement and seven upper labials (Smith 1945; Conant 1950). Values are means ± SD. ADS, anterior dorsal scale; U/S, upper labial scale; NA, not available.

Table 3. Pairwise F_{ST} between the Luther Marsh, Sarnia, and Windsor areas.

	Luther Marsh	Sarnia	Windsor
Luther Marsh	—	—	—
Sarnia	0.204	—	—
Windsor	0.145	0.054	—

Sarnia region or Luther Marsh (Table 4). Luther Marsh snakes had the highest private allelic richness (2.1 ± 0.7 , mean ± SE) followed by Windsor snakes (1.3 ± 0.4) and then Sarnia snakes (0.4 ± 0.3) (Table 4).

Log-likelihood values from STRUCTURE increased as K increased, reaching a plateau at $K = 5$ before decreasing again (Fig. 2). Inspection of the individual assignment probabilities at $K = 5$ revealed a high level of admixture and no individual could be assigned with a high probability (>80%) to the fifth cluster (Fig. 3b). When $K = 4$, 26% of the individuals could not be assigned with a high probability to one of the four clusters, whereas 74% of the individuals could be assigned to one of four clusters with >80% probability (Fig. 3a). Importantly, when K was increased from 3 to 4 in STRUCTURE, there was still a large number of individuals with an assignment probability of greater than 98% in the new cluster (Fig. 3a). In GENELAND, the modal value of the 50 independent runs was 5 (19 occurrences) in the correlated allele frequency model. The correlated allele frequency model in GENELAND has been shown to perform poorly in estimating the K parameter when it is unknown, in particular it tends to overestimate K . Therefore, we also ran a second analysis with the same parameter set but using the uncorrelated allele frequency model. The modal value of K in the 50 runs of the uncorrelated allele frequency model was 4, with 38 independent runs with a posterior density of four independent clusters. Importantly, GENELAND identified the same set of clusters as STRUCTURE with Sarnia and Luther Marsh remaining as distinct genetic clusters (Figs. 3a, 3b, 4). Both GENELAND and STRUCTURE identified snakes from Luther Marsh as having high pairwise F_{ST} values between all other populations (GENELAND = 0.13–0.20; STRUCTURE = 0.18–0.20), whereas snakes from the Sarnia region exhibited more moderate divergence between the Windsor populations. Snakes from the Windsor region exhibited much more complex genetic structuring and seem to group into anywhere from two to three distinct clusters ($K = 4$ and $K = 5$, respectively). The specific boundaries of the genetic sub-structure in the Windsor area are not yet clear.

Discussion

Our study is the first to characterize the population genetic structure of the endangered *T. butleri* across its Canadian range. We show that all snakes collected in Canada contain a single mtDNA haplotype. This haplotype clusters with *T. butleri* from Michigan. Previous authors have proposed that a closely related species, *T. brachystoma*, exists in Luther Marsh; however, our assignment test results indicate that Luther Marsh represents a genetically distinct population of *T. butleri*. The Sarnia and Windsor regions also appear to be genetically distinct populations within the Canadian range and the Windsor region may further be subdivided into possibly three or four subpopulations that show a north to south cline. However, clustering results in the Windsor region are ambiguous and we caution making inferences on the spatial boundaries of possible genetic sub-structure in this region.

Does *T. brachystoma* exist at Luther Marsh?

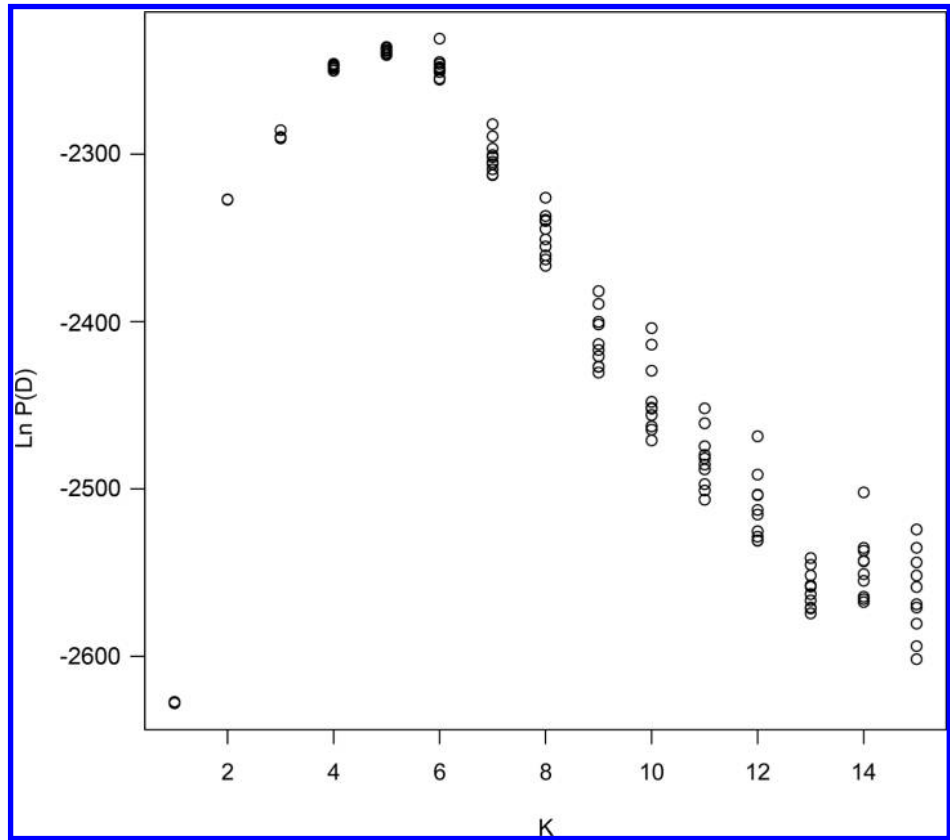
We find no evidence for the existence of *T. brachystoma* in Canada. This conclusion is supported by the presence of a single mtDNA haplotype across Ontario and the lack of two nuclear genetic clusters in the Luther Marsh population.

Table 4. Sample sizes (*N*), mean observed and expected heterozygosities (*H*_O and *H*_E, respectively), allelic richness, private allelic richness, and the mean number of alleles across seven loci in each of the three geographically isolated areas of Butler’s Gartersnake (*Thamnophis butleri*) in Canada.

Population	<i>N</i>	<i>H</i> _O	<i>H</i> _E	Allelic richness	Private allelic richness	No. of alleles
Luther Marsh	23	0.67±0.05	0.69±0.03	5.6±0.8	2.1±0.7	5.71±2.13
Sarnia	24	0.64±0.02	0.72±0.01	5.8±0.4	0.4±0.3	6.0±1.0
Windsor	54	0.69±0.02	0.77±0.01	7.6±0.6	1.3±0.4	9.0±2.0

Note: Values are means ± SE.

Fig. 2. Inferring the number of genetic clusters from a plot of likelihood values for each successive change in KMAX using the admixture model in STRUCTURE version 2.3.1 (Pritchard et al. 2007).



In a recent biogeographical study of the *T. radix* group, Placyk et al. (2012) identified divergent *T. brachystoma* and *T. butleri* mtDNA haplotypes with *T. brachystoma* haplotypes differing by seven mutational steps from the closest haplotypes of *T. butleri* and *T. radix*. Furthermore, the ND2 gene region for *T. butleri* in Ontario clustered with sequences taken from *T. butleri* in Michigan and Ohio rather than *T. brachystoma* from Pennsylvania (Placyk et al. 2012), suggesting that all specimens are *T. butleri*. Conversely, if our sample included only hybrids and no pure *T. brachystoma* (intermediate forms of *T. butleri* in Luther Marsh could represent hybrids between *T. butleri* and *T. brachystoma*; Schueler and Westell 1975), then mtDNA alone may not support the absence of the latter species. When introgressive hybridization occurs between two closely related species, the mtDNA haplotype of one species can become fixed in the closely related sister species and this lack of variation limits one’s ability to identify sympatric sister species using mtDNA alone.

Provided that *T. butleri* and *T. brachystoma* maintain some reproductive isolation, we would expect the nuclear DNA of both species to be different. If this were the case, we would predict that there would be two genetically distinct, syntopic clusters, which would imply reproductive isolation between these groups (i.e.,

one nuclear cluster being *T. butleri* and a second *T. brachystoma*) (Noble et al. 2010). Furthermore, these genetic clusters should correspond to the different “diagnostic” morphological traits between the two species with intermediate forms potentially representing hybrids and having equal probability of clustering to each of the two separate genetic groups (Noble et al. 2010). This predicted correspondence was not upheld at any value of *K*, suggesting that the Luther Marsh population is simply an isolated, genetically divergent population of *T. butleri* that exhibits convergence in scale counts with *T. brachystoma* from Pennsylvania and New York. Alternatively, it is possible that extensive introgression between *T. butleri* and *T. brachystoma* may have led to our inability to detect distinct genetic clusters if there are weak reproductive barriers between these species and they have persisted in this area for some time.

Convergence in morphological traits can evolve in different species owing to similar selective environments (Losos 2011) or could simply be a result of random genetic drift (Stayton 2008; Losos 2011). Convergence in *T. butleri* and *T. brachystoma* is likely due to their similar genetic background and their shared pleiotropic, epistatic, and developmental pathways (Losos 2011). Indeed, *T. brachystoma* and *T. butleri* both prefer open prairie habitat and

Fig. 3. Individual assignment plots from STRUCTURE for populations of Butler’s Gartersnakes (*Thamnophis butleri*) in Ontario. (a) $K = 4$; (b) $K = 5$. Each vertical bar represents a single individual, while each colour corresponds to a single “population” or “cluster”. The y axis represents the probability of an individual being a member of a particular coloured genetic cluster.

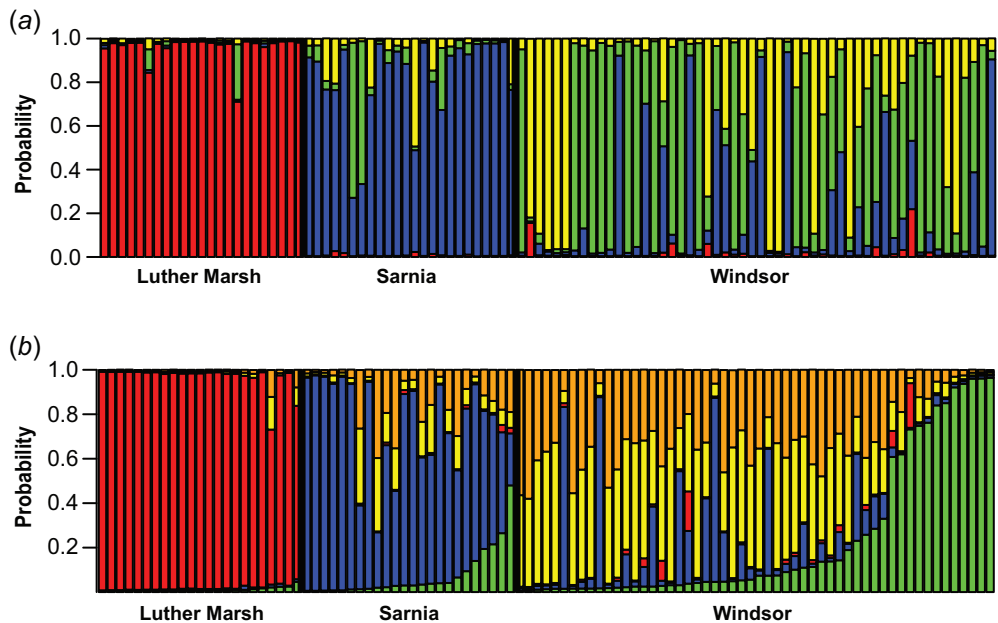
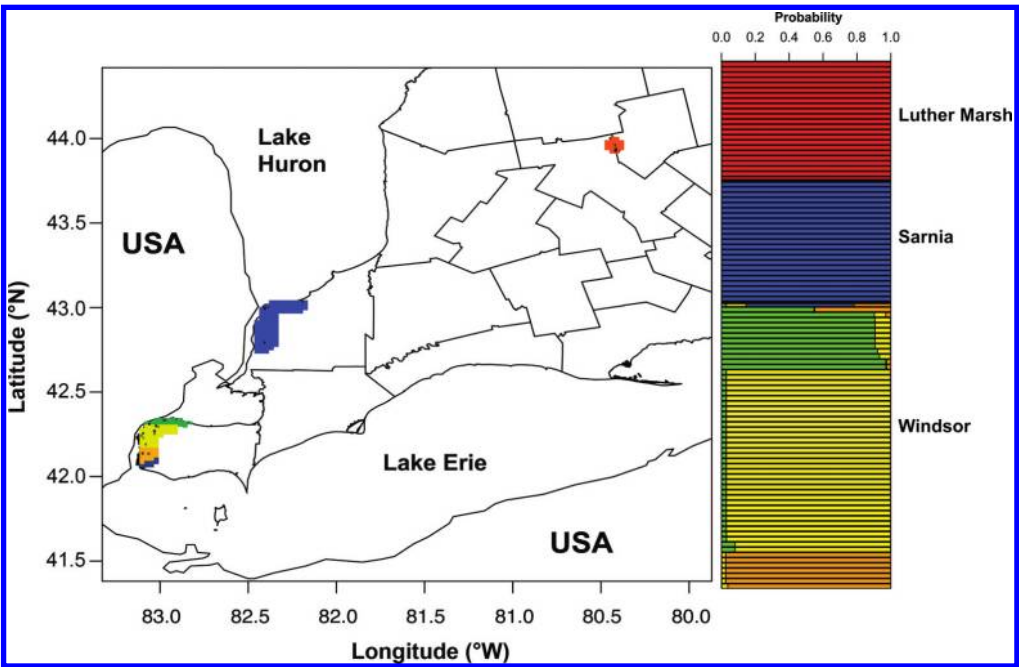


Fig. 4. Map showing the spatial genetic clustering of Butler’s Gartersnakes (*Thamnophis butleri*) across Ontario, Canada. The side-bar graph shows the individual assignment probabilities for each individual in the analysis and the colours on the graph match up with the three general populations (Luther Marsh, Sarnia, and Windsor). Analyses were run in GENELAND (Guillot et al. 2005b).



are dietary specialists on leeches and earthworms (Casbourn et al. 1976). However, it is not clear why *T. butleri* in Luther Marsh would exhibit such a high frequency of convergence to a *T. brachystoma* phenotype compared with samples from the rest of Ontario.

Using scale counts to differentiate between these two very similar species will thus be misleading. Given *T. butleri* and *T. brachystoma* are allopatric, geography is a reasonable indicator of species identity; however, where the latter is brought into question, genetic analysis is necessary.

Genetic structure and conservation of *T. butleri* in Canada

The lack of mtDNA variation in *T. butleri* across Ontario is not particularly surprising given the recent colonization of this area after the glacial retreat approximately 10 000 years before present (Anderson and Lewis 1992; Webb et al. 1993). A similar pattern has been documented in Eastern Foxsnakes (*Pantherophis vulpinus* (Baird and Girard, 1853)) across Ontario in which 700 base pairs of the cytochrome *b* region yielded no nucleotide variability (Row et al. 2010). Low genetic diversity in mtDNA loci has also

been reported from many temperate region vertebrates (Amato et al. 2008 and references therein), again reflecting the relatively recent colonization of these areas after the glacial retreat.

At the nuclear genetic level, Canadian *T. butleri* seem to be composed of at least four to five genetically distinct units. The Luther Marsh population is genetically distinct from snakes in the Sarnia and Windsor regions and maintains a high pairwise F_{ST} with both populations. It is not surprising that Luther Marsh snakes exhibit unique genetic structure, as this population is separated by approximately 250 km from the closest known population in the Sarnia region. An abundance of private alleles and a strong tendency for individuals to “cluster” in the assignment analyses, suggests that this population contains an important component of the genetic diversity for this species in Canada. As a result, the extent of occurrence, habitat availability and quality, as well as management regime, are important considerations for the conservation of this population.

At present, Luther Marsh is a relatively large, contiguous wildlife management area (5680 ha, almost 1/3 of which is a reservoir) managed and owned by a conservation authority and government ministry (Tupman 2010). The known extent of the Luther Marsh population has expanded through recent survey work (COSEWIC 2011) and currently there is an abundance of grassland habitat, some of which is actively managed through controlled burns (Tupman 2010). Additional grasslands are being created through farmland restoration (Tupman 2010). Management and restoration activities are primarily targeted toward waterfowl, wild turkey, and pheasants, but the design and implementation of a monitoring plan for herpetofauna has been recently proposed as a high-priority action (Tupman 2010). Although there is no formal management plan for *T. butleri*, we suggest that given Luther Marsh's unique genetic standing, a formal plan should be made a priority in future management drafts.

The Sarnia *T. butleri* also represent a single genetic cluster (with no clear substructure) and are genetically distinct from snakes in the Windsor area. Furthermore, there are no known extant populations of *T. butleri* between these two areas. As a result, we suggest that snakes from the Sarnia area (Lambton Co.) should be considered a separate population and thus any management plan for this species in the Sarnia area should take this into consideration.

Windsor *T. butleri* appear to be composed of anywhere from two to three genetically distinct populations. GENELAND suggests a series of 2–3 populations between the Town of Amherstberg and the City of Windsor/Town of Tecumseh. This could represent a cline of isolation with distance in this area because the clusters grouped out fairly progressively along the Windsor range. Although there appears to be a north–south cline, we refrain from making inferences on the exact boundaries of these genetic clusters owing to our small samples from many sites and the possibility of uncollected populations in this region, which could contribute to greater genetic subdivision. Nonetheless, the fact that we did observe genetic substructure in this area is interesting and contrasts with fine-scale genetic studies on other snake species in this region. Barriers to dispersal in *P. vulpinus* in southern Ontario seem to be associated with the presence of agricultural fields or in some cases large highways (Row et al. 2010). Surprisingly, fox snakes in the Windsor region appear not to be genetically differentiated even though they inhabit a complex matrix of urban areas composed of small remnant patches of habitat (Row et al. 2010). Similar patterns have been observed in the Eastern Gartersnake (*Thamnophis sirtalis sirtalis* (L., 1758)) at an even larger spatial scale in southwestern Ontario, where there was no evidence of clear population genetic substructure in this region (DiLeo et al. 2010). These studies are in contrast to the Windsor area *T. butleri*, which appear to show evidence for fine-scale genetic subdivision. Such patterns could be due to the differences in dispersal tendencies between *T. butleri* and *T. s. sirtalis* and *P. vulpinus* or it may

suggest that we are witnessing divergence among these populations as a result of small population sizes and (or) anthropogenic barriers to dispersal. More thorough sampling in the future will be necessary to resolve the number and location of genetically distinct populations in the Windsor area. This study provides the basis for such an investigation.

Ambiguity in assignment tests for *T. butleri* in the Windsor region

Making inferences on the ideal clustering solution (K) can be a remarkably difficult task (Guillot et al. 2009), particularly when genetic clusters are weakly differentiated (Manel et al. 2005). STRUCTURE and GENELAND produced slightly different results and uncertainty remains regarding the exact number of clusters in the Windsor region. STRUCTURE and GENELAND both suggested greater admixture within individuals even though the number of genetic clusters (K) estimated did not reflect individual membership to these clusters. The discrepancy within and between programs surely reflects the different modelling approaches and the similar posterior probability between models with $K = 4–6$ (Fig. 2), but may also reflect the small data set that we have, as is common when working with endangered species. Coulon et al. (2006), in their analysis of roe deer (*Capreolus capreolus* (L., 1758)) population genetic structure, found that GENELAND was more sensitive in detecting fine-scale population genetic structure compared with STRUCTURE for which they attributed this difference to use of spatially explicit models and this may explain the slightly larger number of clusters being estimated by some GENELAND runs in our data. However, differences in the level of admixture within individual *T. butleri* and the differences between the programs may also reflect the complex history of *T. butleri* in this region and the low genetic differentiation between some populations.

Colonization of the Canadian region probably occurred within the last 8 000 – 10 000 years and *T. butleri* were likely more widespread and abundant during this time due to widespread presence of suitable habitat (tallgrass prairie) (COSEWIC 2011). This habitat does go through successional changes into forest, which in combination with anthropogenic barriers (Row et al. 2010; COSEWIC 2011) may have increasingly isolated populations very recently, allowing them to diverge to varying degrees. This would generate a complex genetic pattern, which would make the assignment of individuals more difficult because it would generate varied degrees of relatively subtle genetic structure. Furthermore, incomplete sampling of contemporary populations, either because they are unknown or because they have been extirpated as a result of the major changes in land use over the last century, may also be obscuring genetic patterns and making assignment of individuals more difficult. More thorough and complete sampling of Windsor and Sarnia with the aim of also acquiring more individuals per site will likely yield more fine-scale structure in these areas.

Conclusions

In summary, we show that *T. brachystoma* is most likely not present in Canada and therefore morphological variation within the Luther Marsh population likely stems from convergence in these meristic traits between *T. butleri* and *T. brachystoma*. *Thamnophis butleri* appears to be broken into five genetically distinct populations, with individuals from Luther Marsh and Sarnia grouping into separate genetic clusters. The number of genetic clusters in Windsor is more complex, but our data suggest that there are at least two to three.

We provide the first range-wide population genetic study of this species in Canada and provide a valuable contribution to help guide basic management decisions for this endangered species. Future work should aim to resolve important genetic divisions that may exist in the Windsor area and seek to identify new locations where *T. butleri* may exist. Such information will be invaluable

able in helping protect genetic diversity in this fragmented and endangered species.

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